



University of Tehran

College of Engineering

School of Electrical and Computer
Engineering (ECE)



Robotics

Homework 3

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Problem 1: 4 DOF SCARA Robot Jacobian (20 points)

A 4 DOF SCARA manipulator is shown in Fig. 1. Find the Jacobian matrix of the robot. Now assume $\theta_1 = 30^\circ$, $\theta_2 = 60^\circ$, and $l_1 = 1$, $l_2 = 1$, $d_4 = 0.5$ m and twist vector is:

$$\begin{bmatrix} 0 \\ 0 \\ 1.3 \\ -1.15 \\ \frac{\sqrt{3}}{20} \\ 0 \end{bmatrix}$$

Find joint velocity vector in this configuration.

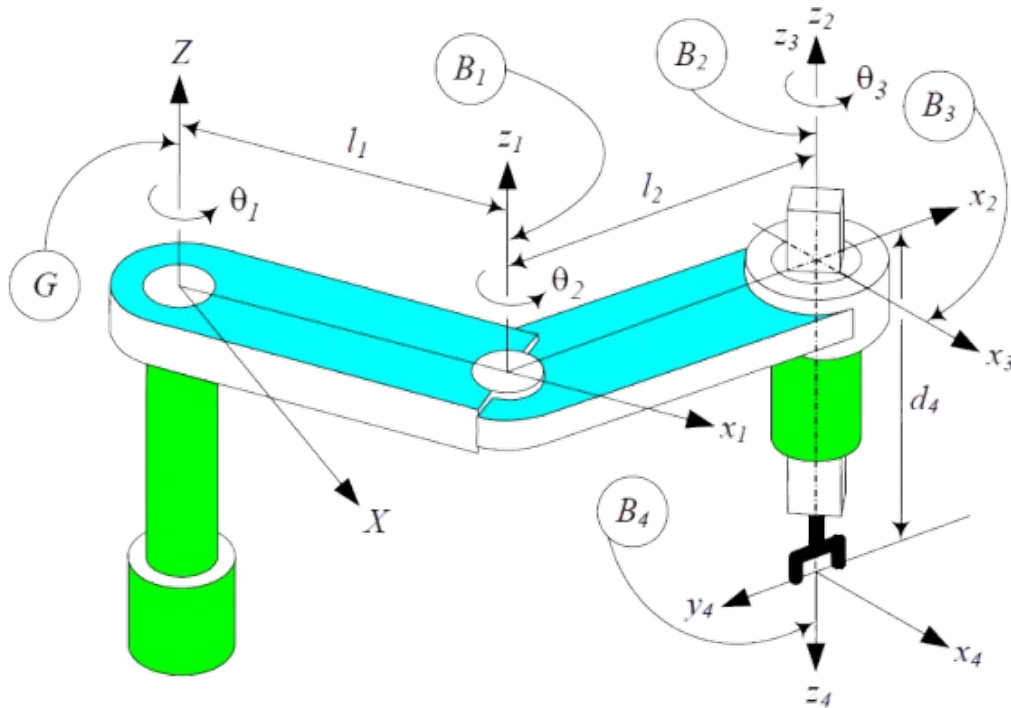


Figure 1: 4 DOF SCARA Robot

Problem 2: PUMA 560 Singularity (25 points)

Fig 2 shows 6 dof PUMA 560 robot which is used in industrial automation.

- Create a table with the Denavit-Hartenberg parameters.
- Use DH table to compute Jacobian matrix of the robot.
- Use Jacobian matrix to find the singular points of the robot and plot them.(you may use MATLAB or Python to find singular points)

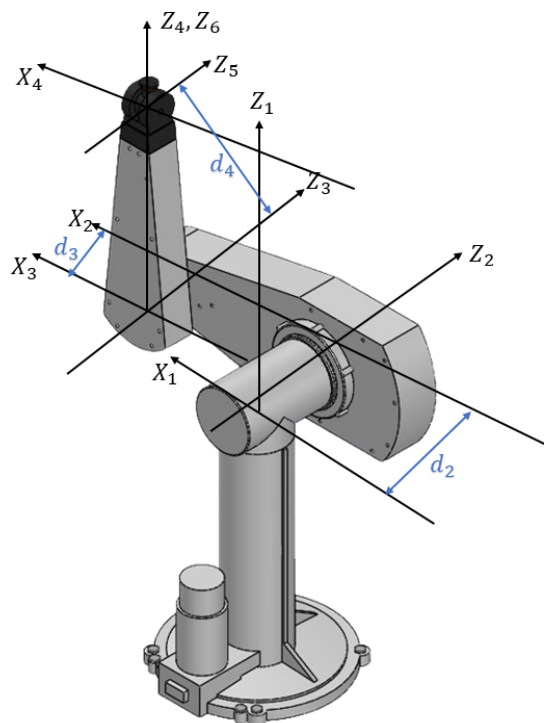


Figure 2: 6 DOF PUMA 560 Robot

Problem 3: Kinematic Sensitivity (15 points)

Kinematic sensitivity refers to the degree to which small variations or disturbances in a robot's joint variables affect the position and orientation of its end-effector. Understanding kinematic sensitivity allows engineers to evaluate how reliable and stable the robot's motion will be in different configurations. Kinematic sensitivity is obtained via the bellow formula:

$$\kappa = \frac{\sigma_{max}}{\sigma_{min}}$$

whereas σ is the singular value decomposition(SVD) on the manipulator's Jacobian matrix. SVD can be derived via the formula bellow:

$$\sigma_i = \sqrt{\lambda_i}$$

λ_i is the eigenvalue of the (JJ^T) matrix. Fig 3 shows a 3R planar manipulator.

- Find the jacobian matrix of the manipulator.
- Now assume l_1, l_2, l_3 are 60, 80, 70 cm and $\theta_1, \theta_2, \theta_3$ are 75, 30, 70 degrees. Find the kinematic sensitivity of the manipulator in this configuration and explain the result.

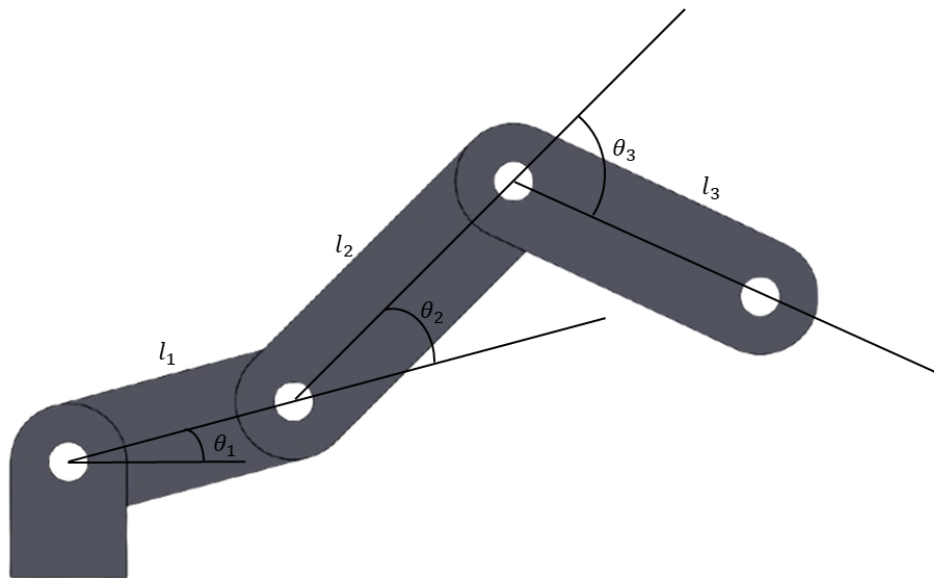


Figure 3: 3R planar manipulator

Problem 4: 3 DOF Robot (25 points)

Fig 4 shows a 3 DOF manipulator in a certain configuration. Assume $l_1 = l_2 = l_3 = l$.

- Find DH parameters of the manipulator. Be precise in drawing the vectors.
- Use DH parameters to solve Forward Kinematic Problem(FKP).
- Solve Inverse Kinematic Problem(IKP) of the manipulator.
- Find Jacobian matrix in general using the DH parameters.
- Find Jacobian matrix in the given configuration.
- Find and plot singular points of the manipulator($l = 80cm$).
- Find the workspace of the manipulator($l = 80cm$).

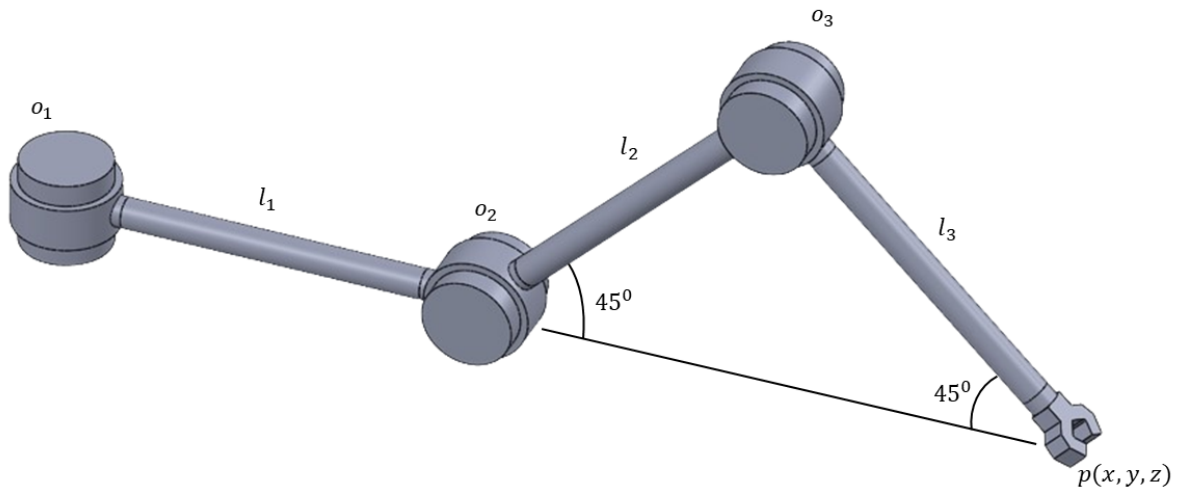


Figure 4: 3 DOF manipulator

Problem 5: Static Of RRPRRR Robot (15 points)

Fig 5 shows a RRPRRR manipulator with its vectors and axis.

- Use the figure to compute the jacobian matrix of the manipulator.
- If $\theta_1, \theta_2, \theta_4, \theta_5, \theta_6$ are 20, 30, 25, 40, 35 degrees, d_2, d_3 are 60, 70cm compute motor torques using the end effector torque and forces given below:

$$\vec{n} = [2, 1, 0.5] \text{ N.m} \quad \vec{f} = [10, 5, 15] \text{ N}$$

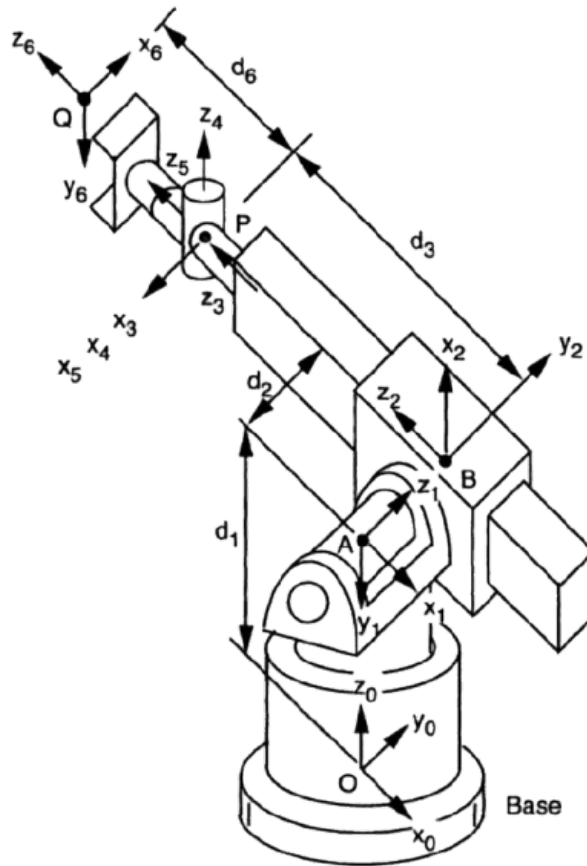


Figure 5: RRPRRR Manipulator

Homework Guidelines and Instructions

- The deadlines are fixed and cannot be changed. If you need extra time, you can use your grace period (14 days) and upload your answers up to 5 days after the deadline.
- If you write your report of this homework in $\text{\LaTeX}$, you will be rewarded 5% extra points.
- Programming problems can either be completed in Python or MATLAB and your codes must be executable and uploaded along with the report.
- This exercise is done by one person.
- If any similarity is observed in the work report or implementation codes, this will be considered as fraud for the parties.
- Using ready-made codes without mentioning the source and without changing them will constitute cheating and your practice score will be considered zero.
- If you do not follow the format of the work report, you will not be awarded the grade of the report.
- Handwritten exercise delivery is not acceptable. Only mathematical procedures and formulas are allowed to be handwritten.
- All pictures and tables used in the work report must have captions and numbers.
- A large part of your grade is related to the work report and problem solving process.
- Please upload the report, code file and other required attachments in the following format in the system: `HW1_[Lastname]_[StudentNumber].zip`
For example: `HW1_Ahmadi_123456789.zip`
- You can ask your questions or doubts either via Telegram group or sending an email directly to the teaching assistants of this homework.
- The questions that are asked in Telegram PV of teaching assistants will not be answered as they are not traceable for the chief TA.
- If you wanted to ask your questions or doubts via email, use the following subject: `"Robotics.HW1-[Lastname]_[StudentNumber]"` and make sure to cc the chief TA at shayanahmadi9999@gmail.com:
 - honarbakhsh.mahdi@gmail.com (Mahdi Honarbakhsh)
- Be happy and healthy